

HANDS-ON ML EXPLORATION

Training My First ML Model

A beginner-friendly walkthrough using Google Teachable Machine

TEACHABLE MACHINE

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PROGRAMME

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SUBMITTED FOR

InAmigos Foundation — Task 3

TASK 03

Machine Learning Model Exploration

1. Introduction

This task is a hands-on introduction to Machine Learning: instead of only reading about how models are trained, the goal is to actually build one, using a beginner-friendly platform that needs no coding at all. This makes it possible to directly experience the core ML workflow — collecting examples, training a model, and testing its predictions — in just a few minutes.

The platform chosen for this exercise is **Google's Teachable Machine** (teachablemachine.withgoogle.com), a free, browser-based tool that lets anyone train an image, sound, or pose recognition model using their webcam or microphone.

2. About the Platform: Teachable Machine

WHAT IT IS

A no-code ML training tool

Teachable Machine is a web tool built by Google that trains real machine learning models directly in the browser. It supports three project types: **Image**, **Sound**, and **Pose** recognition, and requires no installation, account, or programming knowledge.

WHY IT SUITS BEGINNERS

Learn by doing, instantly

Every step — collecting samples, training, and testing — happens visually on one screen, with results appearing within seconds. This makes the abstract idea of "training a model" concrete and easy to understand.

01 Setting Up the Project

Choosing a project type and defining the classes

[Describe here: which project type you chose — Image, Sound, or Pose — and what the classes represent, e.g. "Open Hand" vs "Closed Fist," or "Coffee Mug" vs "Water Bottle." Replace this paragraph with 2–3 sentences about your specific choice and why you picked it.]



1. Project type selection (Image → Standard image model)

2. Classes created (Thumbs Up, Open Palm, Background)

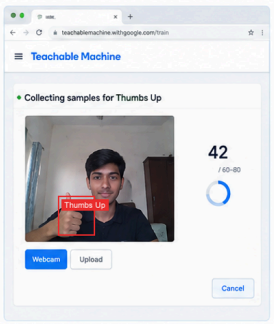
Figure 1: Setting up the project and defining the classes to recognise.

02

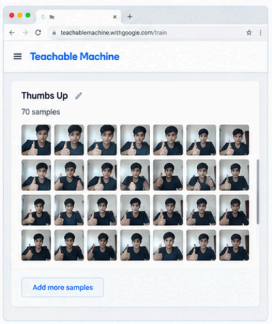
Collecting Training Data

Capturing labelled examples for each class

[Describe here: how many samples you captured per class, and how you introduced variation (different angles, lighting, distances) so the model would generalise well rather than memorise one exact pose or image.]



3. Webcam capture in progress
(for Thumbs Up)



4. Sample gallery
(≈ 60–80 images per class)

Figure 2: Training examples captured live via webcam for each class.

03 Training the Model

Letting the algorithm learn the pattern

[Describe here: what happened when you clicked “Train Model,” roughly how long it took, and what you observed — e.g. an accuracy/epoch graph, or a simple “model trained” confirmation.]

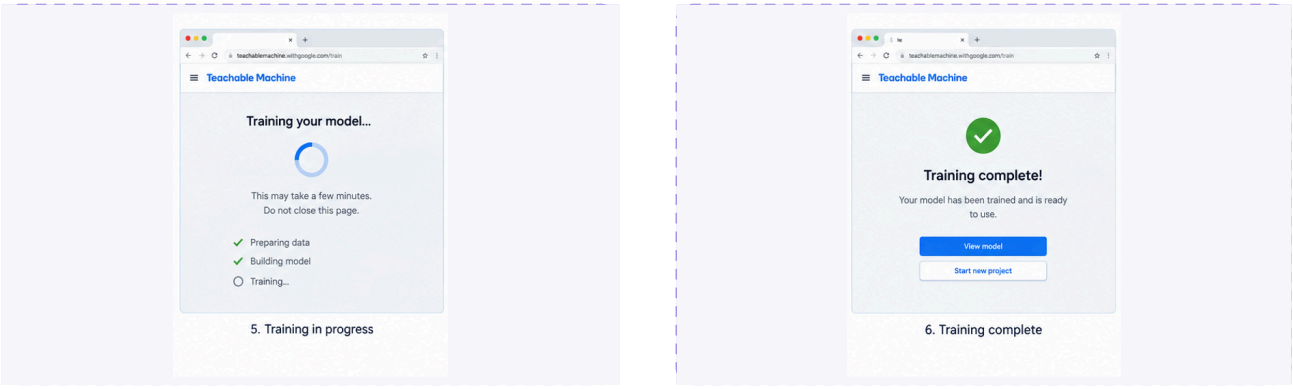


Figure 3: The model training process, from start to completion.

04 Testing the Model

Checking predictions on new, live input

[Describe here: what you tested the model on, what confidence scores it produced for each class, and whether it correctly recognised the object/sound/pose. Mention any misclassifications you noticed and why they might have happened.]

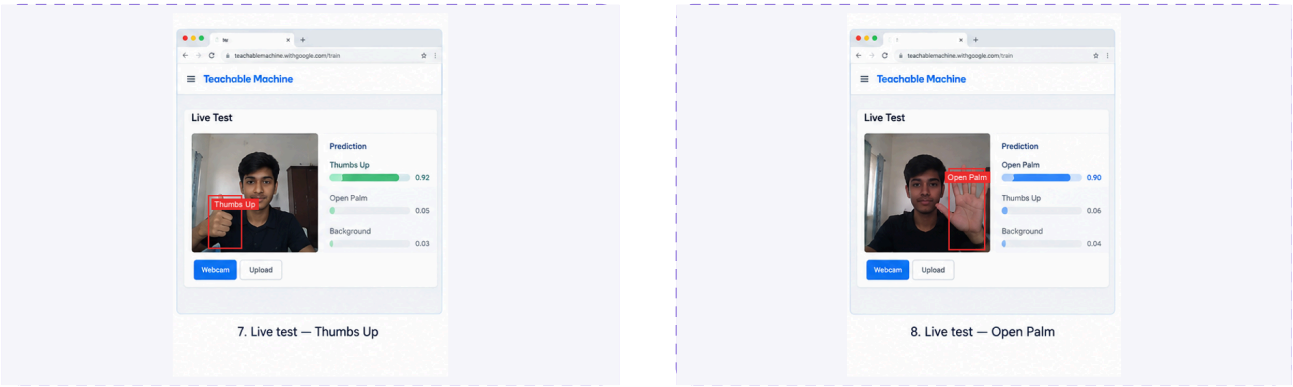


Figure 4: Live predictions from the trained model, with confidence scores.

7. What I Learned

This activity taught me that machine learning depends on the quality and variety of the training data. I learned that a model can be trained without writing complex code by collecting examples, training the model, and testing it with new inputs. I also learned that lighting, camera angle, distance, and background can affect prediction accuracy, so varied training samples are important for reliable results.

8. Conclusion

This project provided a practical introduction to the machine-learning workflow using Google's Teachable Machine. By creating the Thumbs Up, Open Palm, and Background classes, collecting training samples, and testing the model, I learned how the quality and variety of data affect predictions. The model successfully recognised the hand gestures during live testing, although its accuracy could vary with lighting, camera angle, distance, and background. Overall, this activity helped me understand how data collection, training, testing, and evaluation work together to create a useful machine-learning model.